Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/28/2005
PI Name:	Chang, Geoffrey A. Ph.D.		
Project Title:	Crystallization of Integral Membrane Proteins using Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gchang@scripps.edu	**Fax:**	FY 858-784-9985
PI Organization Type:	UNIVERSITY	**Phone:**	858-784-9490
Organization Name:	The Scripps Research Institute		
PI Address 1:	MB-09		
PI Address 2:	10550 North Torrey Pines Road, TPC-20		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92037	**Congressional District:**	49
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	09/01/2001	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Vanderwoerd, Mark J	**Contact Phone:**	256 544-3343
Contact Email:	mark.j.vanderwoerd@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1834		
Performance Goal No.:			
Performance Goal Text:			

We propose to investigate the effects of microgravity for improving the 3D membrane protein crystallization of several transporters that confer drug resistance in the treatment of both infectious diseases and cancer. Our laboratory is determining the high-resolution x-ray structure of two distinct families of mdr transporters that are involved in lipid transport and drug translocation across the cell membrane. We have already produced over twenty distinct 3D crystal forms of these integral membrane proteins using multiple microbial species, different detergents, and crystallization conditions. The production of several mdr transporter crystals is well under way as we are screening more crystallization conditions with several detergents. Microgravity has great potential for creating unique crystal growth conditions that can result in better diffracting protein crystals. In particular, microgravity might offer an opportunity for improving the mosaicities of membrane protein crystals.
The strength of our proposal lies in our ability to investigate the effects of weightlessness on crystal growth of several integral membrane proteins simultaneously. Hence, our membrane protein structure determination strategy will

Task Description:	maximize the probability of obtaining better quality diffracting transporter crystals in space. The 3D crystallization data that will result from this study will provide a large wealth of positive and negative data that will directly benefit the growth of future integral membrane protein crystals using microgravity. A high-resolution crystal structure of mdr transporters will enable us to understand the molecular structural basis of drug resistance in the treatments of infections and cancer. The objectives during the course of our studies on Earth is to: 1. Over-produce and purify several classes of mdr drug transporters for 3D crystallization 2. Crystallize families of mdr transporters and their homologs in various detergents on Earth 3. Crystallize selected classes of mdr transporters using microgravity 4. Study the effects of microgravity on membrane protein crystal growth and determine the x-ray structure of mdr transporters. Multi-drug resistance is a very alarming and a rapidly growing health problem that is a major obstacle in the treatment of both infectious diseases and cancer. The World Health Organization (WHO) has recently reported that mdr bacteria can account for up to 60% of all hospital-acquired infections globally (June 12, 2000 press release). In the United States, for example, 15-20% of patients diagnosed with tuberculosis (TB) are harboring strains of mdr-TB that are resistant to the two most powerful anti-TB drugs, isoniazid and rifampin. Drug resistant bacterial strains that cause gonorrhea, pneumonia, cholera, and malaria are widespread in several third-world countries and are already very difficult to treat with our best front-line drugs. In humans, a similar type of cellular mdr is a major reason for the failure of several chemotherapeutic drugs in the treatment of cancer. Cancer cells that were initially sensitive to a broad range of drugs can frequently reemerge and become resistant to entire families of tumor suppressing drugs. Patients with mdr cancer tumors can often have severe relapses and have few medical alternatives to improve their health condition. A common goal in the chemotherapeutic treatment of both infectious disease and cancer is focused on trying to understand the mechanism by which cells become resistant to a wide-range of drugs and then to find ways to circumvent the cellular mdr. The cause of the mdr phenomenon is the result of the over-production of families of integral membrane proteins that confer multiple drug resistance and function as energy-dependent efflux pumps. These mdr transporters can be classified into two categories that either alter the permeation of drugs through the cell membrane by transporting certain lipids or directly transporting drug molecules out of the cell. Although numerous biochemical and genetic approaches have been used to probe the functional aspects of the multidrug resistance phenotype, there has only been one high-resolution atomic model determined (Chang and Roth, Science, 2001 and Chang, JMB 2003)".
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We propose to investigate the effects of micro-gravity for improving the 3D membrane protein crystallization of several transporters that confer drug resistance in the treatment of both infectious diseases and cancer. Our laboratory is determining the high-resolution x-ray structure of two distinct families of mdr transporters that are involved in lipid transport and drug translocation across the cell membrane. We have already produced over twenty distinct 3D crystal forms of these integral membrane proteins using multiple microbial species, different detergents, and crystallization conditions. The production of several mdr transporter crystals is well under way as we are screening more crystallization conditions with several detergents. Micro-gravity has great potential for creating unique crystal growth conditions that can result in better diffracting protein crystals. In particular, micro-gravity might offer an opportunity for improving the mosaicities of membrane protein crystals. We have completed the crystallization, data collection and data analysis for two membrane-bound drug transporters, MsbA and EmrE. The results of this work were published (Journal of Molecular Biology 330, pp419-30 (2003) and bibliography below). Based on our research results we were able to formulate hypotheses about the molecular mechanism of action for these drug transporters. These mechanisms are an essential stepping stone in the understanding and development of mechanisms to prevent cells from excluding beneficial drugs from their interior cytoplasmic space. The detailed analysis of the molecular mechanisms will ultimately suggest exactly how we may prevent the removal or exclusion of drugs. The amount of detail or resolution of our structures is the most significant parameter in our model. It has been our intention to use the microgravity environment to improve the resolution of the membrane-bound molecular structures, improvement which has been demonstrated in globular (non-membrane-bound) proteins. In this past year we finalized each task objective (listed 1-4 above) with the exception of studying the effects of microgravity on membrane protein crystal growth, because access to microgravity conditions has not been available.
Bibliography Type:	Description: (Last Updated: 09/11/2009)
Articles in Peer-reviewed Journals	Ma C, Chang G. "Structure of the multidrug resistance efflux transporter EmrE from Escherichia coli." Proc Natl Acad Sci U S A. 2004 Mar 2;101(9):2852-7. PubMed PMID: 14970332 Retraction in: Ma C, Chang G. Proc Natl Acad Sci U S A. 2007 Feb 27;104(9):3668. , Mar-2004
Articles in Peer-reviewed Journals	Ma C, Chang G. "Crystallography of the integral membrane protein EmrE from Escherichia coli." Acta Crystallogr D Biol Crystallogr. 2004 Dec;60(Pt 12 Pt 2):2399-402. PubMed PMID: 15583400 , Dec-2004
Articles in Peer-reviewed Journals	Chang G. "Multidrug resistance ABC transporters." FEBS Lett. 2003 Nov 27;555(1):102-5. , Nov-2003